

The Gnomon and the Sun at Its Utmost: An Empirical Tradition of Shadow-Measurement before the Dengfeng Observatory

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Abstract

At Gaocheng, a town in Dengfeng, Henan, a brick terrace stands behind a stone baseline thirty-six metres long. It was built in the Yuan dynasty under Guo Shoujing to measure the noon shadow — and it was placed there because for the two thousand years before him, generation after generation had held that this was 「地中」, the centre of the earth. The phrase is not rhetoric; it is a reading off a ruler: at noon on the summer solstice, an eight-chi gnomon casts a shadow exactly one chi five cun long. This piece sets out the tradition of measurement that leads up to the Dengfeng observatory. It keeps apart the two things the same pole and shadow have been read for, direction and length; traces the term 「日至」, the sun at its utmost, from the oracle-bone inscriptions through the *Zuo Zhuan* to a Western Han calendar manuscript from Hujiacaochang; explains the gnomon (表) and the baseline rule (圭); describes the lacquered pole from Taosi, at 171.8 cm the earliest excavated instrument of the kind; shows that 「地中」 is a reading and therefore moves with latitude; and ends with how a repeated solstice reading yields the tropical year of 365.25 days. Statements are labelled 【通说】 (a general view) or 【一说】 (one view). The piece is descriptive and carries no prediction.

Keywords: gnomon; 日至; solstice; shadow measurement; 地中; Taosi; Dengfeng observatory; tropical year; Han calendar; Chinese calendrics

Introduction

At Gaocheng, a town in Dengfeng, Henan, the ground rises against Mount Song to the north and looks out to Mount Ji to the south, with the Ying River passing the edge of the town. In the

town stands a walled compound of seven successive courtyards; in the innermost of them is a brick structure in the shape of a truncated pyramid, known as the observatory terrace. The terrace is more than ten metres high, and against its northern face runs a stone baseline thirty-six metres long, its surface cut with a water channel and a scale, and still showing the traces of the 「景符」, the shadow-catcher.

The terrace was built in the Zhiyuan period of the Yuan dynasty, under the direction of Guo Shoujing. He built it in order to measure the noon shadow.

Why would a Yuan astronomer go to this small town to measure a shadow? Because for the two thousand years before him, generation after generation had held that this was 「天地之中」, the centre of heaven and earth.

The four characters are not a figure of speech. They are a reading: at noon on the day of the summer solstice, an eight-chi gnomon casts a shadow exactly one chi five cun long. Whoever measured out that number stood at the very centre of the world.

1. Two things to keep apart: the direction of the shadow, and its length

The previous piece in this series was about the lacquer box from the tomb of Marquis Yi of Zeng. It closed with a sentence left hanging: the twenty-eight star names on that box let a person look up and identify the sky, but to fix a date by them one thing is still missing — a way to measure the length of the shadow day by day.

That thing is the gnomon-and-baseline, the 圭表.

Before turning to it, one point has to be prised apart. From one pole and one shadow, the ancients read off two different things.

The first is direction. Look at the tip of the shadow at sunrise, and again at sunset; join the two points and you have true east-west. Both the Zhouli, in the 「Kaogongji」 chapter, and the Zhoubi Suanjing set out this procedure: one pole, one circle, two shadows.

The second is length. Measure the shadow at noon each day, and over a year it first shortens, then lengthens, then shortens and lengthens again, completing one round. That round is the rhythm of the year.

Direction concerns *where*; length concerns *when*. What this section is about is the second.

2. 「日至」: when the sun reaches its utmost

Start with the term. In the old books, 「日至」 means the moment at which the sun reaches its extreme. There are two in a year: one when the shadow is longest, one when it is shortest. On the longest day the sun lies furthest round towards the south, and this is called 「日南至」, the

sun's southern reaching; on the shortest day it lies furthest towards the north, and this is called 「日北至」, the sun's northern reaching.

The later terms 冬至 and 夏至 — winter solstice and summer solstice — grew out of these two words.

「日至」 appears very early. It is already there in the oracle-bone inscriptions. Dong Zuobin devoted a separate volume to it, the *Rizhizhi*, included in his *Yinli pu*, and reported that the inscriptions record 「日至」 in two places: one in the reign of Wu Ding, one in the reign of Wen Wu Ding. No character for shadow is found in the inscriptions, nor any record of shadow measurement; but the two words are there. This point is marked 【一说】: the chronology of the Shang kings is itself still disputed, and the various accounts do not agree on which years these two entries fall in.

By the Spring and Autumn period the records harden. The *Zuo Zhuan*, under the fifth year of Duke Xi, has this line:

春，王正月，辛亥朔，日南至。

The fifth year of Duke Xi of Lu is 655 BCE. Du Yu's commentary on the line reads: 「周正月，今十一月，冬至之日，日南极。」 The first month of the Zhou calendar corresponds to the eleventh month of the Xia calendar, which is the month in which the winter solstice falls.

The *Zuo Zhuan*, under the twentieth year of Duke Zhao, records it once more: 「春王二月己丑，日南至。」 This time the record is a little off: the solstice has landed in the second month. Commentators after Du Yu saw what had happened. The calendar officers of that year had left out an intercalary month; the solar terms shifted with it; and so the transmitted text of the *Chunqiu* declined to write the proper term 冬至, writing only 「日南至」, leaving the wording vague. The matter is recorded in the *Hanshu*, 「*Lüli zhi*」.

These two entries are the earliest solstice records in the transmitted literature. 【通说】

Note one thing: 「日至」 does not refer exclusively to the winter solstice. The *Zhouli*, 「*Diguan · Dasitu*」, has the line 「日至之景，尺有五寸」, and this refers to the summer solstice, because only the summer shadow is short enough to reach one chi five cun. In the same book, the same term falls at the other end.

By the Han, 「日至」 had become a word that could serve as the title of a volume. At the end of 2018, a Western Han tomb, M12 at Hujiacochang in Jingzhou, Hubei, yielded more than four thousand five hundred bamboo slips. Among them were two calendar manuscripts, one inscribed on the reverse with 「历」, calendar, and the other with 「日至」. The one marked 「日至」 consists of one hundred and two slips, recording, for roughly a hundred years from the first year of Emperor Wu's Jianyuan reign (140 BCE), the days on which eight dates should

fall — winter solstice, beginning of spring, spring equinox, beginning of summer, summer solstice, beginning of autumn, autumn equinox, beginning of winter — with the winter solstice placed first. These slips were written before the Taichu li was promulgated: the men of Emperor Wu's time were already using 「日至」 as the name of a calendar manuscript.

Coming upon the two characters 「日至」, one should first see which day it means.

3. The gnomon and the baseline

Now to the equipment.

The instrument for measuring the shadow was called 圭表. Two characters, two objects.

The pole set upright on level ground is the 表, the gnomon. The rule laid flat, marked with a scale and used to measure the length of the shadow, is the 圭, the baseline.

Measuring at noon makes sense. Within a day the shadow is shortest at noon, and that shortest point is easy to catch; at noon the shadow's direction is truest, falling due north; and around noon the length changes most slowly, so that within the same quarter of an hour there is little difference, and the reading does not easily slip.

The Zhouli, in the 「Kaogongji · Jiangren」 chapter, sets out the whole procedure in some detail: 「水地以县, 置槲以县, 眡以景。为规, 识日出之景与日入之景……」 Level the ground, raise the pole upright with a plumb line, and then look at its shadow. This passage is about fixing direction, not length.

How tall is the gnomon to be? The old books give a fixed number: eight chi.

The Zhoubi Suanjing opens with this line:

周髀长八尺, 夏至之日晷一尺六寸。髀者股也; 正晷者句也。

「周髀」 means the gnomon; 「髀」 is the longer leg of a right triangle, 「晷」 the shadow. An eight-chi pole casts, at noon on the summer solstice, a shadow of one chi six cun.

The number eight chi was in use for an extremely long time. It is eight chi in the pre-Qin mathematical books; it is eight chi in the Han commentaries; and not until the Northern and Southern Dynasties did anyone first alter it. Used for so long, it became the standard. The old books give no reason for it.

4. The lacquered pole from Taosi

All of the above is what the texts say. Is there anything under the ground?

There is, and it is much earlier than the texts.

In the autumn of 2002, at the Taosi site in Xiangfen, Shanxi, archaeologists came upon a royal tomb within the walled settlement, numbered II M22. In the south-eastern corner of the chamber stood a wooden pole.

The outer layer of the pole is lacquer. The whole pole was first given a coat of black lacquer, then lacquered in segments of mineral green, and each end of every green segment was lacquered with a band of pink, separated from the black ground. Seen whole, the pole carried a pattern of alternating black and green.

Its surviving length is 171.8 cm. The lower end is well preserved; the upper end was damaged in cleaning, and a stretch of 26.4 cm in the middle has rotted away without trace. In the side niche of the same tomb lay a jade cong and a jade qi, read by the researchers as counterweight and shadow-catcher, that is, as attachments.

The pole carries one particular mark: at 39.9 cm measured from the lower end, a mark has been made. The Taosi chi of the time was about 25 cm, and 39.9 cm converts to exactly one chi six cun.

One chi six cun is the 「夏至之日晷一尺六寸」 of the *Zhoubi Suanjing*, the summer-solstice shadow of one chi six cun.

From the same tomb came a second pole for setting upright: a wooden pole painted all over in red, which on the Taosi chi would stand at about nine chi; in use it was driven one chi into the ground, leaving eight chi above. Eight chi, the height of the gnomon, again matches.

This group of objects the researchers judge to be the earliest archaeologically attested instrument of the gnomon-and-baseline type, dated to roughly 4300–4000 years ago. 【通说】

One layer has to be made clear. To say that a gnomon rule was excavated, or that this is the world's earliest instrument of the kind, is one thing; to say that people four thousand years ago used it to determine the winter solstice is another, and there the claim must be left open. Not one character on the lacquered pole says what it was for. Whether it is a shadow-measuring rule is a judgement assembled from its form, its scale and the objects buried with it, not something the object states of itself. 【一说】

On 21 June 2009, the summer solstice, researchers at the Taosi site made a full-size replica of the excavated instrument and took a measurement, obtaining the noon shadow length for that day. What this step shows is that the method works and that the object can be matched to it; it cannot, in reverse, prove that people four thousand years ago did the same thing with it.

The 1.6 chi on the Taosi pole shows one further thing: the summer-solstice shadow is not a single number valid for the whole country. It varies with latitude.

5. 「地中」 is a reading off a ruler

Now 「地中」 can be discussed.

The *Zhouli*, 「Diguan · Dasitu」, contains the earliest occurrence of the term:

以土圭之法，测土深，正日景，以求地中。日南则景短多暑，日北则景长多寒，日东则景夕多风，日西则景朝多阴。日至之景，尺有五寸，谓之地中。

The 「土圭」 is the rule used to measure the shadow. 「测土深」 means measuring how far north or south the ground lies. Correct the noon shadow with the earth rule, and seek out 「地中」. If the shadow falls short, towards the south, the ground is hot; if it falls long, towards the north, the ground is cold; if it is east or west, the morning and evening shadows are wrong too.

The standard comes down to a single sentence: at noon on the summer solstice, a shadow one chi five cun long, and this is the centre.

Zheng Xuan, annotating the book in the Eastern Han, quotes his elder Zheng Zhong, and puts it more plainly:

土圭之长，尺有五寸。以夏至之日，立八尺之表，其景适与土圭等，谓之地中。今颍川阳城地为然。

An eight-chi gnomon, and at noon on the summer solstice a shadow exactly equal to the one-chi-five-cun earth rule: such a place, it was then held, lay in Yangcheng in Yingchuan commandery. Yangcheng is present-day Gaocheng in Dengfeng.

Look back at the two numbers:

	Basis	Summer-solstice noon shadow
Taosi (southern Shanxi)	the scale on a lacquered wooden pole	one chi six cun
Yangcheng (present-day Gaocheng, Dengfeng)	Zheng Xuan's commentary on the <i>Zhouli</i>	one chi five cun

The difference is one cun. That cun comes out of the difference in latitude. Taosi lies in southern Shanxi, at a higher latitude than the Luoyang region; the noon sun stands lower there, and the shadow is naturally longer.

The word 「地中」 sounds as though it points to a place; in fact its criterion is a reading. When the reading changes, the place moves with it.

Later men saw through this layer. Sima Guang of the Northern Song drew a *Rijing tu*, a chart of the sun's shadow, and Wang Yinglin's *Kunxue jiwen*, juan 9, quotes it: 「日行黄道，每岁有差，地

中当随而转移。故周在洛邑，汉在颍川阳城，唐在汴州浚仪。」 The path the sun travels is itself changing, and 「地中」 moves with it: under the Zhou it was Luoyi, under the Han Yangcheng in Yingchuan, under the Tang Junyi in Bianzhou.

If 「地中」 follows the reading, then measuring 「地中」 was not only astronomy but also politics. He Nu, who has long directed the Taosi excavations, has a phrase for it, four characters: 王者逐中, the king pursues the centre. When the regime changes and the capital moves, the standard of the 「中」 on the gnomon changes with it.

The 「中」 of 「中国」 has its source here. The He zun inscription of the early Western Zhou contains the line 「余其宅兹中国」, the earliest occurrence of the two characters 中国 yet found.

Two points should be marked.

【通说】: the geographical marker of the idea of 地中 came in later ages to be fixed broadly on the Mount Song and Yi-Luo region.

【一说】: Feng Shi holds that as early as 6500 years ago 「地中」 already lay in the Taosi region, since the great tomb at Xishuipo in Puyang, which lies on roughly the same parallel as Taosi, has much the same summer-solstice shadow. This rests on correlating latitude with site, and is not direct evidence.

6. From one 「日至」 to the length of the year

A single 「日至」 written down is only a date. It is by recording them in series that a year comes out.

The difficulty lies at the start. The moment of the winter solstice does not always fall at noon. It may fall at any hour of the day. To pin it down, there is no way but to measure day after day and close in on it.

Measuring on and on, the ancients found something. Liu Zhao's supplement to the Hou Hanshu, 「Lüli zhi」, puts it clearly:

日发其端，周而为岁，然其景不复，四周千四百六十一日，而景复初，是则日行之终。以周除日，得三百六十五四分度之一，为岁之日数。

The sun sets out from its starting point and comes round to make a year, yet the shadow does not return to its place. Only in the fourth year does the shadow fall back on its starting point. Between the two lies one thousand four hundred and sixty-one days. Divide by four, and a year is three hundred and sixty-five and a quarter days.

That quarter-day remainder is where the name 「四分历」, the quarter-day calendar, comes from. The extra whole day is the intercalary day.

By the end of the Spring and Autumn period the length of the tropical year was set at 365.25 days. Against the currently accepted value of 365.2422 days, it is longer by 0.0078 of a day, about eleven minutes a year.

Eleven minutes a year sounds like nothing. Accumulated over sixteen hundred years it comes to ten days. But that is a later story.

What matters is the character of this step. The number was read off the shadow, one day at a time.

7. Where the ruler came to rest

Back to the terrace at the beginning.

Yangcheng, present-day Gaocheng, became the default location for 「地中」 once Zheng Xuan's Eastern Han commentary on the *Zhouli* named it. Later measurements were mostly made there.

In the fourth year of the Yifeng period of Emperor Gaozong of the Tang (679 CE), the taichang boshi Yao Xuanbian was ordered to set up an eight-chi gnomon at the Yangcheng shadow-measuring platform, following the old method; the shadow measured at noon on the summer solstice was one chi five cun, as in the old method. In the eleventh year of the Kaiyuan period of Emperor Xuanzong (723 CE), the taishijian Nangong Yue had a stone gnomon carved there as well. Under Yangcheng county, Henan fu, the *Xin Tangshu*, 「Dili zhi」, has the line 「有测景台」, there is a shadow-measuring platform; that is this matter.

Further on, in the Yuan, Guo Shoujing directed a calendar reform and, for the sake of a 「四海测验」, a survey of the four seas, set up twenty-seven observation points across the country. The one at Dengfeng is the observatory terrace.

The observatory terrace is an enlarged gnomon-and-baseline. The truncated-pyramid body is the gnomon, the long stone baseline beneath it is the rule, and the dimensions are several times the old standard. The enlargement has a reason: the taller the gnomon, the longer the shadow, and the finer the instant represented by the same one-cun division of the scale. The 「景符」 on the terrace body was there to deal with the blurred edge of the shadow cast by a tall gnomon — a thin bronze plate with a small hole in the middle, drawing the edge of the shadow into a single fine line, so that the reading could be exact.

From the 1.7-metre lacquered pole of Taosi to this terrace over ten metres high at Dengfeng, more than three thousand years lie between. What is measured has not changed: the shadow at noon.

The empirical tradition before the Dengfeng observatory is the stretch described here.

The 「中」 that people in Henan speak of can be traced back to this: one reading, measured for two thousand years.

Note on the text

This is a popular exposition, and it makes no claim to original research. The astronomical constants cited are standard values. This English version is issued alongside the Chinese original as the same article in another language. The author's name is 邝楚嘉, Chujia Kuang; he writes under the pen-name 嘉言一得 (Jiayan Yide).

Chinese linear measures are left in their original units. The 尺 (chi) of the Taosi period was about 25 cm; the 寸 (cun) is one tenth of a 尺.

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